Where Do User Ideas Go? Tracing Feature Requests Across Moodle

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Abstract

As open-source software (OSS) adoption continues to grow, effectively managing feature requests across diverse stakeholders becomes critical for sustaining evolution and user satisfaction. Feature requests often originate in community forums but rarely link to issue trackers (~3.5%), creating a significant traceability gap. Users primarily contribute ideas in forums, while developers dominate issue trackers, with limited overlap between the two groups. As requests move across channels, they become more technical and refined, indicating a transformation of user intent. However, this transition is largely ad hoc, with unclear ownership and limited tool support, resulting in weak feedback loops and reduced visibility. Overall, this disconnect hinders effective coordination between users and developers.

Contribution

(1) An empirical quantitative analysis of cross-channel trace links between feature requests discussed in Moodle's community forum and feature requests in Moodle's Jira issue tracker
(2) Semi-structured interviews with Moodle developers to understand how feature requests are received, clarified, prioritized, and linked across different channels
(3) Semi-structured interviews with Moodle community forum participants to understand how they submit feature requests, engage with trace links when they exist, and experience acknowledgment, clarification, and feedback from developers

Problem Statement

Open-source software relies on user feedback, with feature requests often starting in forums and moving to issue trackers. However, these **channels are weakly connected**, making it unclear how **requests transition**, **who manages them**, and **how well user intent is preserved**. This **lack of traceability** leads to **fragmented communication**, **poor visibility**, and **weak feedback loops for both users and developers**.

Communication Channels

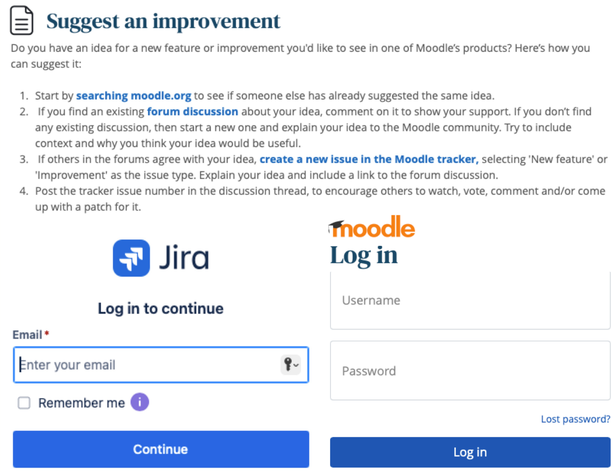
Community Forum — COMMUNITY

⚠ Traceability Gap

Bugzilla · Jira · GitHub

Issue trackers

Moodle: Submit Feature Request

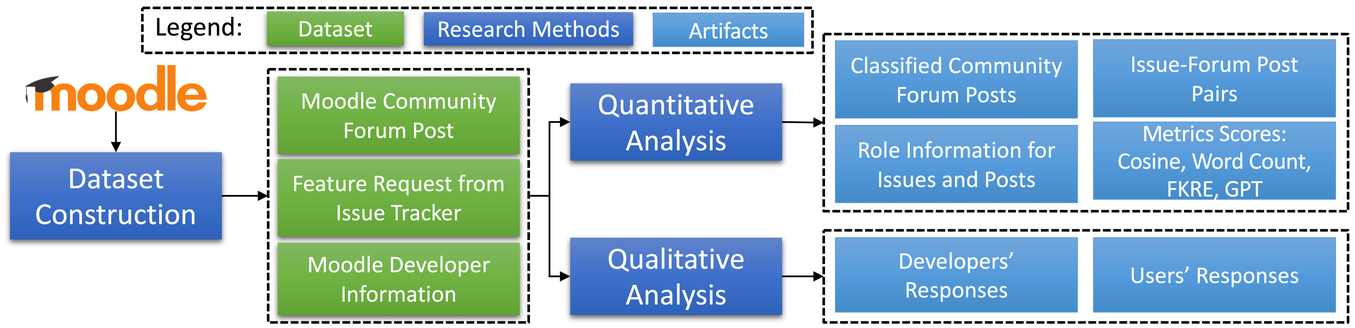


! Confusion for users since multiple channels introduced !

Why it matters?

🔍 **Loss of user intent** during transition across channels
🔗 **Poor traceability** between forums and issue trackers
📉 **Limited visibility** into request status and progress
💬 **Weak feedback loops** between users and developers

Approach Overview



Experiment Design

Quantitative Study		Qualitative Study
Forum Post Classification	Traceability Link Analysis	Moodle Developer Interview
Authorship/Role Analysis	Text Comparision Analysis	Moodle User Interview

Dataset Details

Community Forum	Issue Tracker
204,290 forum posts	23,169 feature related issues

Community Forum Name	# of Posts	
	Total	Sample
generalHelp	70446	500
generalDeveloperForum	23813	500
quizAndQuestionBank	17269	500
themes	10638	500
generalPlugins	4866	500
…		
Other (52 forums – 76900 posts – 21213 samples)		
…		
poodLL	159	88
journal	129	85
artificialIntelligence	41	19
moodleWorkplace	31	19
userExperience	29	15

Moodle Community Forum Roles

Developer	Maintainer			User
Core developers	Moodle workplace	Peer Reviewers	Moodle Cloud	Particularly helpful moodlers
Plugin developers	Plugin guardian	Document Writer	Translator	
	Moodle HQ	Tester		No roles

Results: Quantitative Analysis

- 🔗 **Only ~3.5%** of issues link to forum posts
- 👤 **Forums:** mostly users
- 🤖 **Trackers:** mostly developers
- 🔄 **Requests change across channels**
  •Forum → simpler, user-friendly
  •Issue → more technical, structured
- 📈 **Moderate similarity (0.66)**
  → content is **rewritten, not copied**
- 📖 **Forum posts easier to read** (FKRE score)
  •Forum: ~59 (easier)
  •Issue: ~41 (harder)
- ⚙️ **Issues more technically complex**
  •Forum: 2.77 (easier)
  •Issue: 3.24 (harder)

Results: Qualitative Analysis

- **Developers rely on issue trackers**
  → Rarely monitor forums directly
- **Requests are vague → require interpretation**
  → Developers reformulate user ideas
- 🧩 **No clear process or ownership**
  → Transition from forum → issue is **ad hoc**
- 💬 **Limited communication with users**
  → Feedback loops are weak
- 😕 **Users experience confusion & low visibility**
  → Unclear what happens after submission

🗣️ **What participants said:**
User: "I don't know how it works - the structure is unclear.", "For feature requests, they are not very responsive."

Conclusion & Future Work

✓ Traceability between forums and issue trackers is minimal, leading to loss of user intent and weak collaboration.
- Build automated linking tools and improve workflows to better connect user feedback with development.